

1. Concern 1 : Separate Login Pages for Three User Types.

1.1 Home Page Design:

- **Background:** Use an attractive light blue color with a beautiful light blue theme (Not using any image to attached for project structure).
- **Icons:**
 - **Voter Registration Icon:** Design an attractive icon for user registration (Not using any image to attached for project structure).
 - **User Login Icon:** Design an attractive icon for user login (Not using any image to attached for project structure).
 - **System Admin Icon:** Design an attractive icon for system admin access (Not using any image to attached for project structure).

1.2 Voter Registration Icon Action: When the Voter Registration icon is clicked, it should load the Voter Registration UI.

1.3 User Login Icon:

- **Action:** When the User Login icon is clicked, it should load the User Login UI. Always display a back button to return Home Page again.

1.4 User Login UI:

- **Categories:** The User Login UI should include two categories:
 - **Voter Login UI**
 - **Election Officer Login UI**

1.5 Voter Login Icon

- **Action:** When the Voter Login icon is clicked in the User Login UI, it should load the Voter Login UI. Always display a back button to return User Login UI.

1.6 Election Officer Login Icon:

- **Action:** When the Election Officer Login icon is clicked in the User Login UI, it should load the Election Officer Login UI. Always display a back button to return User Login

UI.

1.7 Election Officer Login UI:

1.7.1 Login Fields: The Election Officer Login UI should display mandatory fields to type the Email Address and Password only.

- **Email Address** (mandatory)
- **Password** (mandatory, not visible)

In election offices are approved by system admin, their email addresses are valid for signup as election officer. So, in signup option of election officer should include only approved voter's email and create password and confirm password with 8 characters. Do not conflict logging credentials with user roles such as voter and admin. As election officer, enter email address and password. Please show both forget password option and signup option. If new password should be replacing the existing password well and Login password should be continued with new password after replacing the existing password. Also update in database in properly.

Duplication Detection is only limited in Election Officer Login Email information field only under Election Officer Login details. Also, Same Email Address could be used in other user type such as Voter. That data does not identify as duplications for voter registration and voter login.

1.7.2 Login Validation:

- Election Officer will log in using their registered Email address and Password.
- If the email or password is incorrect or not matching, display a mandatory message.
- After successfully entering email and password, click login button to navigate the Election Officer Dashboard UI.

1.8 System Admin Icon;

- **Action:** When the System Admin icon is clicked in home page, it should load the System Admin UI.

1.8.1 Super Class Name and Password:

- The system admin will use a predefined super user email and password to access the system admin UI.
- Please implement simple login in System Admin UI to enter predefined super user email and password to log and do other functions well.

1.8.1.1 System Admin Login Validation:

- If the email or password is incorrect or not matching, display a mandatory message.
- Also, please give simple change password option to reset the predefined password properly. If new predefined password should be replacing the existing predefined password well and Login password should be continued with new predefined password after replacing the existing predefined password. Also update in database in properly.
- After successfully entering email and password, click login button to navigate the System Admin UI.

2. Concern 2 : Election Officer Sign Up Process.

2.1 Election Officer Sign Up from Voter Side

Registered Voters can request themselves as Election Officer through the self-request option in the voter dashboard. The request should be sent for system admin for final approval. After approving by system admin, requested voter can act as active election officer with respective role in election officer UI. If not system admin approves the request as election officer, can request again it. Option for self-unenroll as election officer who was previously as election officer also. Also, if voters are acting as an election officer in charge of ongoing election, they are unable to self-unenroll as election officers until elections are entirely finished. So, please pop up message regarding that to understand for voter properly while clicking self-unenroll.

2.2 Election Officer Sign Up from Election Officer Side

In election offices are approved by system admin, their email addresses are valid for signup as election officer. So, in signup option of election officer should include only approved

voter's email and create password and confirm password with 8 characters. Do not conflict logging credentials with user roles such as voter and admin. As election officer, enter email address and password. Please show both forget password option and signup option. If new password should be replacing the existing password well and Login password should be continued with new password after replacing the existing password. Also update in database in properly.

Duplication Detection is only limited in Election Officer Login Email information field only under Election Officer Login details. Also, Same Email Address could be used in other user type such as Voter. That data does not identify as duplications for voter registration and voter login.

3. Concern 3 : The Process of Adding Election Positions and Assigning Candidate to Those Positions (Both Voter Dashboard and Election Officer Dashboards).

3.1 Create Election Process Should be correct as below (Implementation should be launch on Election Officer Dashboard UI and Voter Dashboard UI)

- **Phase 1: Election Creation and Nomination Setup (Election Officer)**

When an election officer begins creating a new election, the process follows this sequence:

Step 1 - Initial Election Setup: The election officer creates the basic election structure through their dashboard. At this stage, they define the election positions (such as President, Vice President, Secretary, etc.) and establish a nomination period with specific start and end times. During this phase, they do not yet set the actual voting dates or finalize other election details that come later, after candidates have been confirmed.

Step 2 - Publishing Nomination Period: Once the election officer publishes the nomination period, the system automatically makes these positions visible to all registered or customized voters in their Voter Dashboard. Respective voters can now see which positions are open for nomination and the deadline for self-enrollment. It's important to note that the election officer cannot modify or delete the nomination period once it's published, this ensures fairness and prevents manipulation of the nomination process.

- **Phase 2: Candidate Self-Enrollment (Voter Side)**

During the active nomination period, registered respective voters have specific options available to them:

Self-Nomination Process: Any registered respective voter can nominate themselves as a candidate through the self-enrollment option in their Voter Dashboard. However, there's a critical restriction: each voter may only self-nominate for one position within the same election. If a voter attempts to nominate themselves for multiple positions, the system should prevent this action and display a clear message explaining that they can only be a candidate for one position per election.

Un-enrollment Option: If a voter changes their mind during the nomination period, they should have the ability to withdraw their candidacy through a self-un-enroll option. Once they un-enroll, they could potentially nominate themselves for a different position (still limited to one position total), as long as the nomination period remains open.

Visibility to Election Officer: All self-nominated voters appear immediately in the Election Officer Dashboard under a pending candidates section for the respective election. This allows the election officer to monitor who has nominated themselves in real-time.

- **Phase 3: Candidate Review and Assignment (Election Officer)**

After the nomination period closes automatically at the specified end time, the election officer moves into the candidate review phase:

Review Authority: The election officer reviews each self-nominated voter and makes one of two decisions for each nominee:

First, they may approve the nomination, which officially assigns that voter as an eligible candidate for their chosen position. The system should update the voter's dashboard to show their "Approved" or "Confirmed Candidate" status for that position.

Second, they may reject the nomination, which means that voter will not appear as a candidate option when voting begins. The system must clearly display the "Rejected" or "Not Approved" status in the voter's dashboard, so they understand they were not selected as an eligible candidate. This transparency is important for voter awareness.

Automatic Ineligibility: If a voter did not self-nominate during the nomination period, they are automatically ineligible to be a candidate for any position in that election. The election officer does not need to take any action for these voters; they simply never appear in the candidate pool.

- **Phase 4: Final Election Configuration (Election Officer)**

Once all candidate assignments are complete, the election officer proceeds to finalize the election:

Voter Assignment: The election officer selects which registered voters are eligible to vote in this election. Only these assigned voters will see the election appear in their Voter Dashboard under the "Ongoing Elections" section when voting begins.

Final Details Entry: Now the election officer enters the complete election details including the election name, official election date, voting start time, voting end time, and any other relevant information. This is also when the election officer confirms all positions and their respective approved candidates.

None of the Above Option: For each position in the election, the system must automatically include a "None of the Above" option as a default choice. This allows voters to indicate they do not support any of the available candidates for that particular position. The election officer does not need to manually add this option; it should be built into the system for every position automatically.

- **Phase 5: Publishing and Conducting the Election**

Publishing Restrictions: When the election officer publishes the finalized election, the system locks all election details permanently. No editing is permitted after publication, not the positions, not the candidates, not the voting times, not the eligible voters. This immutability ensures election integrity.

Election Visibility: Once published and when the voting start time arrives, the election automatically appears in the "Ongoing Elections" section of the Voter Dashboard for all eligible voters who were assigned by the election officer. These voters can now cast their votes for each position, including selecting the "None of the Above" option if they choose.

The election remains visible and ongoing (ongoing elections) in the Election Officer Dashboard as well, allowing the officer to monitor the election status until the voting end time is reached. Display real-time voting statistics and progress in very simply and attractively. Do not reveal voters' votes for candidates at any time.

3.2 Created Ongoing Elections – Election Officer Dashboard View

3.2.1 Created Ongoing Elections – Election Officer Dashboard View

The Election Officer Dashboard provides a comprehensive real-time view of all elections that have been created and are currently ongoing. This section should be embedded directly within the main Election Officer Dashboard UI, not as a separate page.

Real-Time Statistics Display:

For each ongoing election, the Election Officer Dashboard shall display real-time voting statistics and progress information. This information must be presented in a visually appealing and easily understandable format. The statistics should include metrics such as the total number of eligible voters assigned to the election, how many of those voters have already cast their votes, the percentage of voter turnout, and similar aggregate information.

It is critical that while these statistics are displayed, the system shall never reveal which specific candidates any individual voter chose. The statistics should only show aggregate participation data, not voting preferences. The principle of secret ballot must be maintained at all times, with no information displayed that could allow the election officer to determine how any specific voter voted.

Automatic Election Conclusion:

When the designated voting time period for an election ends, the system shall automatically close the election without requiring any manual intervention from the election officer. The election should transition from an "ongoing" state to a "completed" state, and voting should be immediately disabled for that election.

System Continuity During Officer Logout:

Similar to the voter requirements, the system must handle election officer logouts gracefully. When an ongoing election is published and active within its designated time period, the election officer may log out from the system at any time. This logout action shall have absolutely no impact on the ongoing election. The election shall continue to operate normally, voters shall be able to access the election and cast their votes without any disruption, vote counting shall proceed correctly, and the election shall complete at its scheduled end time exactly as planned. The election execution is completely independent from whether the election officer is currently logged in.

Dashboard Stability:

A critical requirement is that the Election Officer Dashboard UI must remain stable and error-free at all times. The dashboard simultaneously displays multiple elements including the interface for creating new elections, the real-time statistics and progress of all currently ongoing elections, and the embedded PDF results reports for completed elections. Despite this complexity, the dashboard shall never crash or become unresponsive.

The system must be designed to handle these multiple embedded components efficiently. Creating a new election should not interfere with the display of ongoing elections. Viewing ongoing election statistics should not conflict with displaying completed election results. All components should coexist seamlessly within the dashboard interface. If any component encounters an error, that error should be handled gracefully and contained to that component without bringing down the entire dashboard. The Election Officer should always be able to access the dashboard functionality regardless of what is happening with individual elections.

3.3. Voter Interface Requirements in On Going Election UI

3.3.1 On-Going Election Icon – Voter Dashboard

When the voter clicks the "On-Going Election" icon from their dashboard, the system shall load and display the On-Going Election UI dashboard described in sections of 3.3.3.1 and 3.3.2.

3.3.1.1 On-Going Election UI - Main Display – Voter Dashboard

The On-Going Election UI serves as the main interface where voters can view and participate in elections that are currently active and for which they have been assigned eligibility by an election officer.

Display Logic:

When a voter accesses this screen, the system shall evaluate which elections to display based on the following criteria. First, the system checks if there are any elections that have been created, are currently ongoing within their designated voting time period, and for which this specific voter has been assigned and marked as eligible by an election officer. If such elections exist, they shall be displayed in the interface with all relevant details including the election name, remaining time for voting, and the positions available for voting.

If there are no elections matching these criteria, the system shall display a clear message stating "No ongoing election available at this moment" to inform the voter that they currently have no elections to participate in.

There is a special case to consider when some elections have finished while others remain ongoing. If at least one election that the voter previously had access to has completed, but other elections are still active, the system shall display a banner message at the top of the screen. This banner should read "Previous results can be viewed in the Results section" and should provide guidance that finished election results are accessible through the Results icon on the voter dashboard. This banner should not interfere with the display of any ongoing elections that are still available for voting.

Navigation Behavior:

The navigation behavior requires careful attention to maintain data integrity. When the voter clicks the back button to return to the Voter Dashboard, the system behavior depends on the voter's current state. Under normal circumstances when the voter is simply viewing the On-Going Election UI without being in the middle of an active voting session, clicking the back button should simply return them to the Voter Dashboard.

However, there is a critical requirement when the voter is actively engaged in voting. If the voter has begun the voting process for an election and has started entering their candidate selections, but then clicks the back button before submitting their votes and completing the random number verification, the system must handle this carefully. In this scenario, the system shall immediately clear all data that the voter has entered for their current voting session. This includes any candidate selections they made for any positions. The voter should be returned to the Voter Dashboard, and importantly, their voting state for that election should be reset to "not voted" so they can start fresh if they choose to vote again later. This ensures that partially completed votes are never accidentally submitted and that voters always have full control over their final submission.

System Continuity During Voter Logout:

An important system behavior concerns what happens when a voter logs out while an election is ongoing. The system must be designed to handle this gracefully. When an ongoing election is published and active within its designated voting time period, a voter may log out from the system at any time. Whether the voter has already cast their vote or has not yet voted, their logout action shall have no impact whatsoever on the ongoing election. The election shall continue to run normally, other voters shall be able to access and vote without any disruption, and the election shall complete at its scheduled end time regardless of individual voter login states. The system must ensure that voter authentication state is completely independent from election execution state.

3.3.2 Voting Process - Detailed Flow – Voter Dashboard

The voting process involves several carefully orchestrated steps to ensure security, authenticity, and a smooth user experience.

Time Display:

When a voter enters an election to cast their vote, the system shall prominently display the remaining time available for voting in this election. This countdown should be clearly visible throughout the voting process so the voter knows how much time remains before the election closes.

Candidate Selection:

The voter shall be presented with all positions in the election, and for each position, they can view the candidates and select their preferred candidate. The interface should be intuitive and clearly show which selections have been made. After the voter has made their selections for all positions they wish to vote in, a "Submit" button shall become available or remain prominently displayed.

Random Number Verification Process:

When the voter clicks the "Submit" button, they are not immediately voting. Instead, the system enters a verification phase to confirm the voter's identity. At this point, the system shall generate a random five-digit number. This random number must be sent to the voter's registered email address that is associated with their account.

It is important to note that this email functionality should be implemented as simply as possible. The system should use a straightforward email sending mechanism that does not incur costs or require complex third-party API integrations. The goal is to send a basic email containing the random five-digit number to the voter's email address on record.

After triggering the email, the system shall present the voter with an input field where they must enter the five-digit random number they received via email. The system shall validate the entered number against the generated number.

Verification Outcomes:

If the voter enters the random number correctly, the system shall accept their vote submission, encrypt (Simple Hash Encryption) the voting data securely to prevent any data breaches, and officially record their vote in the database. The system shall then display a confirmation message stating "You have Voted" and provide additional information that results can be viewed through the Results icon once the election concludes.

If the voter enters an incorrect random number, the system shall reject the submission and display an error message stating "Unable to submit due to wrong random number." The voter should be given another opportunity to enter the correct number.

Time Constraints on Random Number:

The random number verification is subject to time constraints to maintain security. Once a random number is generated and sent to the voter's email, it shall remain valid for exactly 15 minutes. A countdown timer should be displayed to the voter showing how much time remains for them to enter this random number.

If the 15-minute period expires before the voter successfully enters the random number, the system shall invalidate that random number. The voter may then request a new random number, which will generate a fresh five-digit code and send it to their email, starting a new 15-minute countdown.

However, there is an important overall constraint to consider. The voter can only request new random numbers while the election's overall voting period is still active. If the election is scheduled to end in 10 minutes, for example, and the voter requests a random number with only 8 minutes remaining, they will receive the random number but will only have 8 minutes to use it before the election closes, not the full 15 minutes. The system should prevent random number requests if there is insufficient time remaining for a meaningful verification attempt before the election closes.

Post-Voting Behavior:

Once a voter has successfully submitted their vote after proper random number verification, the system shall display the confirmation message "You have Voted" along with guidance that "Results will be available in the Results Icon." After a brief moment to allow the voter to read this confirmation, the system shall automatically return the voter to the On-Going Election UI. If there are other elections still ongoing that the voter has not yet voted in, those elections should still be visible. The election the voter just completed should either be removed from the list or clearly marked as "Voted."

Duplicate Voting Prevention:

The system must implement strict controls to prevent any voter from casting more than one vote in the same election. Once a voter has successfully submitted their vote for an election after completing the random number verification, their voting status for that election shall be permanently recorded as "voted." Any subsequent attempts to access that election for voting

should be blocked, and the voter should see a message indicating they have already voted in this election.

Election Conclusion:

When the designated voting time period for an election concludes, the election shall automatically close regardless of whether individual voters have cast their votes. If a voter is logged in and viewing the On-Going Election UI when an election ends, that election should be removed from their list of available elections. The voter should be able to continue viewing any other ongoing elections normally. The system should handle the election closure gracefully without causing any errors or requiring voters to refresh their screens.

Once an election concludes, the final results shall be automatically compiled, encrypted securely, and published to the Results section accessible via the Results icon on the voter dashboard. Voters should be able to view these results at their convenience.

3.3.3 Results

- Allow downloading reports as PDFs.
- Final Election Results PDF:
Include the number of assigned and eligible voters with employees Numbers, the number of voters who voted with employees' numbers, vote counts for each candidate, and the winner candidates with their images (If equal candidates for same position, it should be also displayed). Display data with attractive graphs, tables, and pie charts. Do not reveal voters' votes for candidates at any time.
- Final Election Result PDF is named as the created election name.
- Ensure that the Final Election Result PDF is visually appealing and easy to understand. Please give customize option to download and visible.